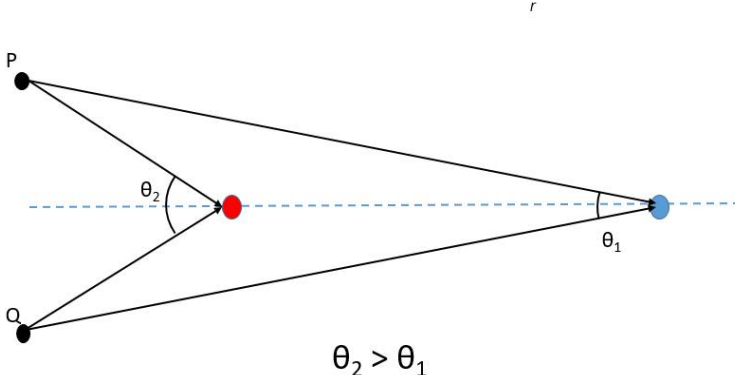


Qns	Answer	Marks
3(a)	limit for which light reflecting off the 2 features can be <u>just distinguished</u> when <u>first minima</u> of the diffraction pattern of one feature coincides with <u>central maxima</u> of the diffraction pattern of the other feature .	B1 B1
3(b)(i)	<u>diffraction pattern</u> formed when light from horizontal scan lines enters through pupil <u>angular separation</u> between lines <u>increases</u> when distance to TV screen decreases, making the two diffraction patterns distinguishable  $\theta_2 > \theta_1$	B1 B1
3(b)(ii)	Distance between horizontal scan lines, $s = \frac{27.9 \times 10^{-2}}{720} = 3.88 \times 10^{-4} \text{ m}$ Diameter of pupil, $b = 3.00 \times 10^{-3} \text{ m}$ Applying Rayleigh's criteria $\frac{s}{r} = \theta \approx \frac{\lambda}{b}$ $r = \frac{sb}{\lambda}$ $= \frac{(3.88 \times 10^{-4})(3.00 \times 10^{-3})}{400 \times 10^{-9}}$ $= 2.91 \text{ m}$	C1 M1 A1
3(b)(iii)	b increases when pupil dilate, applying Rayleigh's criteri (minimum resolvable angle decreases), r increases	M1 A1

Qns	Answer	Marks
4(a)(i)	electromagnetic radiation produced whenever charged particle is	B1